Cyber Security Task 2 (2026)

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Phishing Email Detection & Awareness System

OVERVIEW

Phishing is a **digital trap** where a criminal pretends to be someone you trust—like your bank, Netflix, or even your boss—to trick you into giving up private information. Instead of "hacking" your computer with complex code, they use **social engineering** to hack your emotions. By sending an urgent email or text claiming your "account is locked" or "money is missing," they hope you will panic and click a link. That link leads to a **fake website** that looks exactly like the real one. When you "log in" to fix the problem, you are actually typing your username and password directly into the criminal's hands. It is much easier for a thief to trick a person into handing over their keys than it is to break down a digital vault door.

USING HEADER ANALYZER TOOL:

```
Delivered-To: employee@company.com
Received: from mail-gateway.attacker-server.net (unknown [192.0.2.1])
        by mx.google.com with ESMTPS id z62si4321915wmb.22.2026.03.19.10.45.00
        for <employee@company.com>;
        Thu, 19 Mar 2026 10:45:00 -0700 (PDT)
Return-Path: <support-verification@auth-secure-mail.top>
From: "TrustedBank Security" <alerts@trustedbank.com>
Subject: Urgent: Your Account Will Be Locked
Date: Thu, 19 Mar 2026 10:44:55 -0700
Message-ID: <550e8400-e29b-41d4-a716-446655440000@mail.com>
Authentication-Results: mx.google.com;
       spf=fail (google.com: domain of support-verification@auth-secure-mail.top does not designate 192.0.2.1 as permitted sender)
       dkim=none;
       dmarc=fail (p=REJECT) action=none header.from=trustedbank.com
Content-Type: text/html; charset="UTF-8"

<html><body>[The phishing content from the screenshot goes here]</body></html>
```

Delivery Information

- ✖ DMARC Compliant (No DMARC Record Found)
  - ✖ SPF Alignment
  - ✖ SPF Authenticated
  - ✖ DKIM Alignment
  - ✖ DKIM Authenticated

Relay Information

Received Delay:	**0 seconds**

1. Identification of Phishing Indicators

Based on the visual sample and technical header analysis, the following red flags were identified:

- **Mismatched Identity:** The email claims to be from trustedbank.com, but the technical "Return-Path" shows it actually originated from auth-secure-mail.top.
- **Urgency & Fear:** The subject line "Urgent: Your Account Will Be Locked" is designed to make the user panic and bypass caution.
- **Authentication Failure:** Technical checks (**SPF** and **DMARC**) both failed. This is the "digital proof" that the sender is an impostor.
- **Spelling Errors:** Words like "recieved" and "discrepency" are misspelled, which is highly unusual for a professional bank.

2. Email Risk Classification

- **Risk Level: CRITICAL / PHISHING**
- **Reasoning:** The email failed all three major security protocols (SPF, $DKIM$, $DMARC$). It uses a deceptive link and a generic greeting to target the user's credentials.

3. Brief Overview

Think of this email as a **digital trap**. A criminal is pretending to be your bank to trick you into "opening the door" to your account. By telling you there is a problem with your money, they hope you will panic and click the link. That link leads to a fake website that looks real; when you "log in" there, you are actually handing your username and password directly to the attacker.

4. Prevention Tips: Do's and Don'ts

Do	**Don't**
Do hover your mouse over links to see where they actually lead.	**Don't** click links in emails that demand immediate action or "verification."
Do check for spelling mistakes and generic greetings like "Dear Customer."	**Don't** trust an email just because it uses a familiar bank logo.
Do go directly to the bank's official website by typing the address yourself.	**Don't** download attachments or enter passwords on pages linked from an email.

VISUAL ANALYSIS:

TrustedBank™

Dear valued customer of TrustedBank,

We have recieved notice that you have recently attempted to withdraw the following amount from your checking account while in another country: $135.25.

If this information is not correct, someone unknown may have access to your account. As a safety measure, please visit our website via the link below to verify your personal information:

http://www.trustedbank.com/general/custverifyinfo.asp

Once you have done this, our fraud department will work to resolve this discrepency. We are happy you have chosen us to do business with.

Thank you,
TrustedBank

Member FDIC © 2005 TrustedBank, Inc.

1. **Risk Classification**

Risk Level: CRITICAL (Phishing)

2. **Identification of Phishing Indicators (Red Flags)**

Indicator	**Specific Finding from Sample**	**Explanation**

Generic Greeting	"Dear valued customer of TrustedBank,"	Real banks use your legal name.
Spelling Errors	"recieved", "discrepency"	Professional financial institutions use automated spell-checkers and editors thus never show mistakes
Fear & Urgency	"...someone unknown may have access to your account."	Attackers use fear to make you act quickly without thinking or verifying the source.
Suspicious URL	http://www.trustedbank.com/...	The **http** instead of **https**. The S stands for “secure”. A real bank would never process "personal information" over an unencrypted connection.

3. **How the Attack Works.**

This attack uses Social Engineering to bypass technical security.

1. The Hook: The attacker creates a fake "problem" (a $135.25 withdrawal) to cause panic.
2. The Bait: They offer a "solution" (the link) that appears to go to the bank's website.
3. The Catch: If a user clicks the link, they are sent to a fake website that looks like the bank. When the user enters their username and

password to "verify," the attacker captures those credentials in real-time.

4. Prevention & Employee Guidelines

Do’s

- Inspect the Sender: Check the actual email address behind the name "TrustedBank."
- Use the "Hover Test": Hover your mouse over the link to see the real destination URL in the browser's bottom-left corner.
- Go to the Source: If you are worried about an account, open a new browser tab and type the bank's address manually (e.g., www.yourbank.com) or use their official app.

Don’ts

- Don't Click: Never click links in emails that ask for "verification" of personal or financial data.
- Don't Reply: Replying to the email confirms your address is active, leading to more attacks.
- Don't Trust Logos: Attackers can easily copy and paste high-quality logos from the internet.

5. Final Professional Assessment

This email is a high-risk phishing attempt. Even though the link text looks legitimate (trustedbank.com), in a real scenario, the hidden destination would likely be a look-alike domain (e.g., trusted-bank-security.net). The combination of poor spelling and outdated copyright information makes this an easy-to-spot threat for a trained user.

DISCLAIMER

This assessment was conducted on scanme.nmap.org, a server publicly provided by the Nmap Project specifically for authorized scanning practice. No exploitation, brute force, or harmful activity was performed. This report is produced for educational purposes as part of a cybersecurity internship